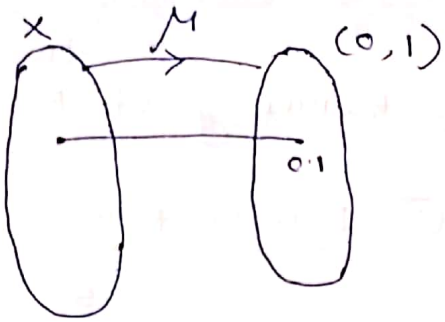

Membership function:-

A membership function of a fuzzy set $\tilde{A}$ is defined as

$$\mu_{\tilde{A}} : X \rightarrow [0,1]$$

$$\tilde{A}(x) = \mu_{\tilde{A}}(x)$$

Representation:-

$$\tilde{A} = \left\{ \frac{\mu_1}{x}, \frac{\mu_2}{y}, \frac{\mu_3}{z}, \ldots \right\}$$

Characteristic of Fuzzy set:-

① Empty Fuzzy set $\rightarrow$ $\tilde{A} = \phi \Leftrightarrow \mu_{\tilde{A}}(x) = 0$

eg:- $\tilde{A} = \left\{ \frac{0}{a}, \frac{0}{b}, \frac{0}{c} \right\}$

② Universal Fuzzy set $\rightarrow$ $\forall\, \mu_{\tilde{A}}(x) = 1$

eg:- $\tilde{A} = \left\{ \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \right\}$

③ Hight of Fuzzy set $\rightarrow$ Highest degree of membership of an elt of any set A

$$h(\tilde{A}) = \sup_{x \in X} \mu_{\tilde{A}}(x)$$

eg:- $\tilde{A} = \left\{ \frac{0.2}{a}, \frac{0.5}{b}, \frac{0.7}{c} \right\}$

$$h(\tilde{A}) = 0.7$$

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④ Equality of two Fuzzy Set:-

ⓐ Equal Fuzzy set:-

$$\tilde{A} = \left\{ \frac{0.5}{a}, \frac{0.7}{b}, \frac{0.9}{c} \right\}, \quad |\tilde{A}| = 3$$

$$\tilde{B} = \left\{ \frac{0.5}{a}, \frac{0.7}{b}, \frac{0.9}{c} \right\}, \quad |\tilde{B}| = 3$$

ⓑ Equivalent Fuzzy set:-

$$\tilde{A} = \left\{ \frac{0.5}{a}, \frac{0.7}{b}, \frac{0.9}{c} \right\}$$

$$\tilde{B} = \left\{ \frac{0.5}{a}, \frac{0.8}{b}, \frac{0.9}{c} \right\}$$

Ex.

$$\tilde{A} = \left\{ \frac{.3}{1}, \frac{.4}{2}, \frac{.6}{3}, \frac{.9}{4} \right\}$$

$$\tilde{B} = \left\{ \frac{.3}{1}, \frac{.5}{2}, \frac{.7}{3}, \frac{1}{4} \right\}$$

$$\tilde{C} = \left\{ \frac{.3}{1}, \frac{.6}{2}, \frac{.4}{3}, \frac{.5}{4} \right\}$$

$$\tilde{D} = \left\{ \frac{.3}{1}, \frac{.6}{2}, \frac{.4}{3}, \frac{.5}{4} \right\}$$

$$\tilde{C} = \tilde{D}$$

⑤ Containment of Fuzzy set:-

$$\tilde{A} = \left\{ \frac{.1}{a}, \frac{.3}{b}, \frac{.5}{c} \right\}$$

$$\tilde{B} = \left\{ \frac{.2}{a}, \frac{.5}{b}, \frac{.1}{c} \right\}$$

$$\tilde{A} \leq \tilde{B} \quad \text{or} \quad \tilde{A} \subseteq \tilde{B}$$

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Fundamental operation on Fuzzy set

- Union
- Intersection
- Complement

Union of Fuzzy set:- Let X be a non-empty set and let $\tilde{A}$ and $\tilde{B}$ be two fuzzy set on X. Then

$$\tilde{A} \cup \tilde{B} = \sup_{x \in X} \{ \mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x) \}$$

$$= \max_{x \in X} \{ \mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x) \}$$

$$= \max_{x \in X} \{ \tilde{A}(x), \tilde{B}(x) \}$$

Ex. Let $X = \{1, 2, 3, 4\}$ and

$$\tilde{A} = \{ (1, .2), (2, .5), (3, .8), (4, 1) \}$$

$$\tilde{B} = \{ (1, .3), (2, .6), (3, .9), (4, 1) \}$$

$$\tilde{A} \cup \tilde{B} = \{ (1, .3), (2, .6), (3, .9), (4, 1) \}$$

Intersection of Fuzzy set:- Let X be a nonempty set and $\tilde{A}$ and $\tilde{B}$ be two fuzzy set on X. Then

$$\tilde{A} \cap \tilde{B} = \cancel{\inf} \min \{ \mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x) \}$$

$$= \inf \{ \mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x) \}$$

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Ex. Let $X = \{1, 2, 3, 4\}$

$\tilde{A} = \{(1, .2), (2, .5), (3, .8), (4, 1)\}$

$\tilde{B} = \{(1, .3), (2, .6), (3, .9), (4, 1)\}$

$\tilde{A} \cap \tilde{B} = \{(1, .2), (2, .5), (3, .8), (4, 1)\}$

Complement of fuzzy set:- The complement of fuzzy set $\tilde{A}$ on X is defined by

$$\boxed{\tilde{A}' = \tilde{A}^c = 1 - \mu_A(x)}$$

Ex. Let $X = \{1, 2, 3, 4\}$

$\tilde{A} = \{(1, .1), (2, .3), (3, .5), (4, 1)\}$

Then

$\tilde{A}' = \{(1, .9), (2, .7), (3, .5), (4, 0)\}$

Normal Fuzzy set:-

A fuzzy set $\tilde{A}$ is said to be normal fuzzy set iff $\mu_{\tilde{A}}(x) = 1$ i.e. $\mu_{\tilde{A}}(x) = 1$ for atleast one $x \in X$

Ex. Let $X = \{1, 2, 3, 4\}$

$\tilde{A} = \{(1, .2), (2, .5), (3, .8), (4, 1)\}$

$\therefore$ $\tilde{A}$ is normal fuzzy set.

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Book:- Fuzzy set and Their Application
~~Sudhir pandey~~ H. J. Zimmermann, springer.

Operation on Fuzzy set:-

① $(\tilde{A} \cup \tilde{B})(x) = \max\{\tilde{A}(x), \tilde{B}(x)\}$

② $(\tilde{A} \cap \tilde{B})(x) = \min\{\tilde{A}(x), \tilde{B}(x)\}$

③ $\tilde{A}^c(x) = \tilde{A}'(x) = 1 - \tilde{A}(x)$

These operations are called standard fuzzy set.

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① **Support of fuzzy set:-** The support of fuzzy set $\tilde{A}$ on X is a classical set which is defined by

$$\text{Support}(\tilde{A}) = \{ x \in X : \mu_A(x) > 0 \}$$

eg:- Let $X = \{1, 2, 3\}$

$$\tilde{A} = \{(1, \cdot 2), (2, \cdot 3), (3, 0)\}$$

$$\text{Support}(\tilde{A}) = \{1, 2\}$$

② **α-cut set or cut worthy set:-**

The α-cut set of fuzzy set $\tilde{A}$ of the set X is a crisp set which is defined

$$^{\alpha}A = \{ x \in X : \mu_{\tilde{A}}(x) \geq \alpha \}$$ (which is crisp set)

~~eg:~~ where $\alpha \in [0, 1]$

eg:- Let $X = \{1, 2, 3, 4\}$,

$$\tilde{A} = \{(1, \cdot 2), (2, \cdot 4), (3, \cdot 6), (4, \cdot 9)\}$$

Find α-cut set for $\alpha = \cdot 3$

Solution:-

$$^{\cdot 3}\tilde{A} = \{2, 3, 4\}, \text{ where } \alpha = \cdot 3$$

Since,

$$\mu_{\tilde{A}}(1) = \cdot 2 < 3$$

$$\mu_{\tilde{A}}(2) = \cdot 4 > 3$$

$$\mu_{\tilde{A}}(3) = \cdot 6 > 3$$

$$\mu_{\tilde{A}}(4) = \cdot 9 > 3$$

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